

Marina

Applicant name: Adrian Peter Radomski
Submitted: Friday, August 29, 2025 kl. 5:08 PM
Created: Friday, August 29, 2025 kl. 5:08 PM
Number of appendices: 6
Funding body: Danmarks Innovationsfond
Instrument: Innofounder/Innofounder
Case number: 5302-00044A

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1. Application

PROJECT TITLE

PROJECT TITLE	
Project title:	Marina
The idea in a nutshell:	Marina is an AI medical assistant for ships. It guides officers through patient assessments in 100+ languages, generates compliant reports, and flags critical conditions - reducing costly evacuations while improving healthcare for 1.89 million seafarers worldwide (https://marinahealth.org).

PREVIOUS APPLICATIONS

PREVIOUS APPLICATIONS	
Have you previously applied for a grant from the Innovation Fund with the same company or idea?:	No

CLASSIFICATION

CLASSIFICATION	
Political theme:	Life science, health and welfare technology
Here, you must describe why your idea fits with the chosen theme. If you have chosen, that your idea does not fit within a theme, then you must describe why your idea falls outside the themes. You can only choose one theme.:	Marina addresses critical healthcare gaps for 1.89 mio. seafarers working far away from medical care. This AI-powered medical assistant enables ship officers with minimal training to conduct structured, multilingual patient assessments and generate compliant medical reports. The solution leverages digital health technologies including voice recognition optimized for maritime environments, AI-powered symptom analysis, and automated red-flag warnings for critical conditions. It operates offline while integrating with existing telemedical infrastructure. Marina directly improves patient outcomes by eliminating language barriers through 90+ language support, reducing diagnostic delays, and preventing unnecessary evacuations costing \$250,000 each. The system transforms how non-medical personnel handle emergencies, providing real-time clinical decision support and standardized documentation. Beyond maritime applications, the technology scales to other remote healthcare scenarios including rural clinics, military medicine, and emergency response. Marina demonstrates how AI can augment healthcare delivery in underserved populations, improving access and quality while reducing costs.
The project's primary Innovation Area:	Computer and information sciences
Primary application sector:	Information og kommunikation

PROJECT DESCRIPTION

ASSESSMENT CRITERIA - QUALITY OF THE IDEA	
PLEASE DESCRIBE YOUR BUSINESS IDEA. WHAT PROBLEM, NEED OR DESIRE DOES YOUR BUSINESS IDEA TRY TO MEET?	
Helptext:	In this section you should furthermore describe: - What is your solution? - How and why does it work?
Please describe your business idea. What problem, need or desire does your business idea try to meet?:	Commercial vessels operate with 1.89 million seafarers without onboard doctors. Minimally-trained officers handle medical emergencies while facing language barriers, producing incomplete reports and delayed diagnoses. This results in 11,000 annual diversions costing \$250,000 each, with 20% preventable. Marina is an AI medical assistant conducting voice interviews in 100+ languages, guiding structured assessments and generating compliant reports with red-flag warnings. Advanced speech recognition captures conversations in noisy maritime environments while AI analyzes symptoms using maritime protocols, ensuring complete data regardless of officer experience or crew language. The system operates offline and integrates with telemedical services. Marina eliminates communication errors, standardizes assessment quality, and provides consistent documentation. This enables faster shore-based medical decisions, reducing unnecessary evacuations while improving seafarer health outcomes.
PLEASE DESCRIBE HOW YOUR IDEA IS INNOVATIVE	
Helptext:	In this section you should furthermore describe: - What is new about your solution? - How it is based on relevant knowledge from your education, research and/or work experience?

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Please describe how your idea is innovative:

Marina is the first maritime-specific AI medical assistant combining multilingual voice processing, AI-powered clinical assessment, and maritime medical protocols. No existing solution addresses comprehensive shipboard medical challenges. The voice-first system eliminates language barriers through real-time processing in 100+ languages while guiding non-medical personnel through structured assessments. Speech recognition functions in noisy engine rooms and harsh deck conditions, converting conversations into standardized reports. Built on emergency medicine expertise from Radio Medical Denmark and health informatics research at SDU, clinical protocols derive from actual maritime cases. Development leverages PhD-level engineering experience. This represents the first conversational AI application to remote maritime healthcare, addressing crew language diversity, communication delays, acoustic interference, and limited connectivity while maintaining regulatory compliance.

PLEASE DESCRIBE HOW YOUR BUSINESS IDEA DIFFERS FROM COMPETING SOLUTIONS, PRODUCTS OR SERVICES

Helptext:

In this section you should furthermore describe:

- Who are your competitors, and how do you differ from them?
- How does this difference create value for your customers vis-à-vis competing solutions?
- How does the competition affect your chances of successfully marketing your solution?
- How do you intend to protect your solution from being copied?

Please describe how your business idea differs from competing solutions, products or services:

Current maritime medical solutions address only partial challenges. TMAS providers like Radio Medical Denmark offer remote consultation but rely on manual documentation without AI support. Ship electronic health records focus on documentation without AI analysis. Maritime medical simulators provide training but no real-time assistance during emergencies. Marina uniquely combines AI-powered symptom analysis, multilingual voice processing across 100+ languages, maritime-specific medical protocols, offline functionality, and automated report generation. Existing competitors lack AI decision support and struggle with language barriers affecting diverse international crews, creating communication gaps that compromise patient safety. Marina eliminates communication errors that compromise patient safety, standardizes assessment quality regardless of officer experience levels, reduces diagnostic delays critical in maritime environments, and prevents costly evacuations averaging \$250,000 each. The system provides legal protection through compliant documentation meeting regulatory standards. Strategic advantages include first-mover positioning in maritime AI healthcare and deep domain expertise from 15 years of maritime physician experience. Exclusive maritime medical data and protocols cannot be replicated by general healthcare AI companies lacking specialized knowledge.

ASSESSMENT CRITERIA - IMPACT

PLEASE RENDER PROBABLE THAT THERE IS A NEED, USERS AND PROBABLY ALSO PAYING CUSTOMERS AND A MARKET OF CONSIDERABLE SIZE - INCLUDING AN INTERNATIONAL ONE, IF POSSIBLE

Helptext:

In this section you should furthermore describe:

- Who are the customers/users, and how do they like your idea?
- Have you sought to identify users and a market for your idea?
- What is the size and potential of this market

Please render probable that there is a need, users and probably also paying customers and a market of considerable size - including an international one, if possible:

The maritime medical market shows substantial need with 1.89 million seafarers working without onboard doctors. Radio Medical Denmark handled 4,829 cases in 2024 with 279 evacuations/diversions, with studies showing 20% are preventable through better initial assessment. Primary buyers are shipping companies seeking to reduce diversion costs (\$250,000 each). Users include ship officers conducting assessments and shore-based telemedical providers. Marina has secured endorsements from captains, corporate medical directors at top-5 shipping companies, and TMAS providers. Selected for Maritime Stars accelerator with backing from Bluetech Center, Danske Rederier, and ShippingLab, currently conducting vessel trials. The global maritime industry operates 50,000+ commercial vessels. Conservative 10% adoption at \$5,000 annual subscription creates a \$25M market, expandable to military medicine, offshore platforms, and remote healthcare applications. Maritime operations are inherently international. Denmark's maritime reputation enables Nordic expansion, then global scaling. Cloud deployment allows rapid geographic expansion without physical infrastructure requirements.

PLEASE RENDER PROBABLE HOW YOU CAN ESTABLISH A FINANCIALLY SUSTAINABLE BUSINESS BASED ON THE IDEA, AND HOW AN INNOFOUNDER COURSE WILL INCREASE YOUR CHANCES OF DOING SO

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TYPE: UFS_ANSOEGNING_IFD_INNOFOUNDER_UK

WORKFLOW ID: 254E1546-1B81-4905-8090-2F026EA87059

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Helptext:

In this section you should furthermore describe:

- Specifically how you plan to make money on the idea and to establish a financially sustainable business that can ensure dissemination of your solution.
- The expected business model, revenue structure and pricing.
- How do you expect to disseminate the solution in the market?

Please render probable how you can establish a financially sustainable business based on the idea, and how an Innofounder course will increase your chances of doing so:

Marina's revenue model uses \$5,000 annual subscriptions for shipping companies and SaaS for smaller operators. Preventing one diversion (\$250,000) covers implementation costs multiple times, creating strong ROI justification. Cost structure includes cloud infrastructure, multilingual speech processing, regulatory compliance, and customer support. The scalable SaaS model delivers high margins after initial development, supporting sustainable growth. Market dissemination follows three phases: Danish maritime companies leveraging existing partnerships, Nordic expansion through Denmark's maritime reputation, then global scaling via direct sales and strategic partnerships. Digital marketing targets smaller operators while industry conferences support enterprise sales development. Financial sustainability shows conservative 1% global vessel adoption generating \$2.5M+ annual revenue. The maritime sector's high diversion costs justify adoption decisions, while subscription-based revenue ensures recurring income and predictable growth patterns. Innofounder provides business model refinement, pricing strategy optimization, enterprise sales process development, and investor pitch prep.

PLEASE RENDER PROBABLE HOW YOUR BUSINESS IDEA HAS THE POTENTIAL TO CREATE VALUE FOR DENMARK IN THE FORM OF FINANCIAL GROWTH AND/OR SOLVING SIGNIFICANT SOCIETAL CHALLENGES

Helptext:

In this section you should furthermore describe:

- What value does the solution create for society, e.g. by solving societal challenges, creating growth and employment, etc.?
- Please describe the quantitative and/or qualitative goals of this value creation and be as specific as possible.

Please render probable how your business idea has the potential to create value for Denmark in the form of financial growth and/or solving significant societal challenges:

Marina positions Denmark as a global leader in maritime AI healthcare by leveraging the country's maritime heritage and technological expertise. The solution addresses healthcare equity for over 20,000 Danish seafarers while reducing maritime emissions through prevented evacuations. Quantitative goals include creating 50+ high-skilled Danish jobs in AI development and maritime health, generating €2M+ annual export revenue within five years, reducing preventable maritime evacuations by 20% globally, and saving the industry \$275M annually through improved medical decisions. Marina reinforces Denmark's maritime innovation leadership, building on the Maersk legacy. The exportable healthcare technology demonstrates Danish commitment to seafarer welfare, enhancing international maritime relationships and potentially influencing global health standards through IMO channels. The first-mover position in maritime AI healthcare creates sustainable competitive advantages for Danish maritime.

ASSESSMENT CRITERIA - QUALITY OF THE EXECUTION

PLEASE DESCRIBE AND RENDER PROBABLE THAT YOUR TEAM HAS THE COMPETENCES TO REALISE THE IDEA

Helptext:

In this section, you should furthermore describe:

- How does the diversity of the team help you realize the idea?
- What are your abilities to promote and support the idea?
- What insight does the team lack to solve future tasks, and how do you plan to act on this?
- Please state to which extent team members who do not apply for admission will be affiliated to activities relating to the business idea during the Innofounder programme (e.g. state the number of hours per week).

Please describe and render probable that your team has the competences to realise the idea:

Our team combines essential clinical and technical competencies for Marina's development. Peter Biesenbach brings 15 years of maritime physician experience with deep understanding of shipboard medical challenges and emergency protocols. Adrian Radomski contributes PhD-level health informatics and AI development expertise from University of Southern Denmark. Team diversity bridges clinical practice and technical implementation. Peter's maritime experience ensures practical relevance and user acceptance, while Adrian's health informatics background addresses regulatory requirements for medical AI systems. Academic affiliation provides research credibility and regulatory pathway support. We have industry connections through Maritime Stars accelerator participation and shipping partnerships. Academic backing enables research publications for credibility building in medical and maritime communities. Missing competencies include business development, enterprise sales strategy. We'll address these through Innofounder mentorship, advisory board recruitment from maritime executives, and potential co-founder addition with commercial experience. Peter will dedicate 20 hours weekly.

PLEASE RENDER PROBABLE THAT YOUR BUSINESS IDEA IS REALISTIC AND CAN BE IMPLEMENTED IN PRACTICE

Helptext:

In this section, you should furthermore describe:

- The status of the development and remaining development.
- How far have you come in realizing the idea?
- Which development tasks must be solved during the Innofounder program.
- Please describe all remaining development tasks and be as specific as possible. Which technical, regulatory, and other tasks have you identified, and how do you expect to address them?

Please render probable that your business idea is realistic and can be implemented in practice:

Marina demonstrates strong implementation feasibility with a working prototype featuring AI functionality, maritime-optimized voice recognition, and 100+ language support. The system conducts structured assessments, generates reports, and flags critical conditions offline. Vessel trials with shipping partners validate real-world performance. Innofounder development tasks include completing MDR 2017/745 compliance, expanding to 100+ languages, optimizing AI accuracy, developing enterprise security, building scalable infrastructure, and creating training modules. Remaining tasks involve enhancing speech recognition for extreme noise, improving AI through expanded data, developing mobile apps, and creating telemedical APIs. Regulatory work includes CE marking, Danish Maritime Authority approval, GDPR compliance, and device classification. Commercial development focuses on pricing validation, sales processes, onboarding workflows, and support infrastructure. Implementation follows controlled trials, parallel regulatory approval, and gradual market expansion based on proven results.

PLEASE RENDER PROBABLE THAT AN INNOFOUNDER PROGRAM CAN HELP YOU PROPEL YOUR BUSINESS IDEA TO A STAGE WHERE IT IS FINANCIALLY SUSTAINABLE

Helptext:

In this section, you should furthermore describe:

- Where are you when the course ends?
- How do you plan to fund the next step?

Please render probable that an Innofounder program can help you propel your business idea to a stage where it is financially sustainable:

Innofounder addresses Marina's key gaps in enterprise sales, regulatory navigation, and investor readiness while leveraging our technical foundation and industry validation. By course end, we expect completed regulatory submission with preliminary approval, enterprise sales pipeline with 3-5 shipping companies in negotiations, validated business model, and investment-ready materials. Funding strategy targets €3-5M Series A from maritime investors and Danish B2B venture capital. Initial customer revenue of €200K-500K bridges scaling operations. Innofounder provides sales development mentorship, regulatory expertise, and investor network access, accelerating go-to-market by 12-18 months. Maritime industry connections facilitate customer acquisition. The structured approach addresses startup challenges while maintaining focus on proven market demand, positioning Marina for sustainable growth in maritime AI healthcare.

ACTIVITY PLAN

ACTIVITY PLAN

ACTIVITY PLAN 1

Activity:

Who does it?:

Result/milestone:

Period:

Regulatory compliance framework development and MDR 2017/745 submission

Adrian with regulatory consultants and Peter for clinical validation

Completed regulatory submission and preliminary approval

Months 1-4

ACTIVITY PLAN 2

Activity:

Who does it?:

Result/milestone:

Period:

Enterprise sales pipeline development and customer acquisition

Both team members with Innofounder mentors and advisors

3-5 shipping companies in advanced negotiations, first paid pilots launched

Months 3-8

ACTIVITY PLAN 3

Activity:

Who does it?:

AI optimization

Adrian with technical team and maritime domain experts

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Result/milestone:	Enhanced AI accuracy and full multilingual support operational
Period:	Months 2-6

ACTIVITY PLAN 4

Activity:	Business model validation and pricing strategy finalization
Who does it?:	Both team members with customer feedback and market research
Result/milestone:	Validated pricing model and sustainable revenue projections established
Period:	Months 4-7

ACTIVITY PLAN 5

Activity:	Investment preparation and Series A fundraising initiation
Who does it?:	Both team members with Innofounder investor network support
Result/milestone:	€3-5M Series A funding secured or advanced term sheet negotiations
Period:	Months 8-12

ACTIVITY PLAN 6

Activity:	Scalable infrastructure development and enterprise security implementation
Who does it?:	Adrian with cloud architecture specialists and security consultants
Result/milestone:	Enterprise-grade platform ready for commercial deployment at scale
Period:	Months 5-10

PARTNER DATA

PARTNER MASTER DATA

Is the idea based in a company?:	Yes
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COMPANY

CVR no.:	45619648
P no.:	1031342771
Name:	Marina Health Denmark ApS
Has revenue been achieved based on the idea?:	No
Has other funding been obtained for the idea?:	Yes

ANDEN FINANSIERING 1

Amount of funding:	50.000,00 kr.
Who has funded:	Mikrolegat

ANDEN FINANSIERING 2

Amount of funding:	430.000,00 kr.
Who has funded:	SDU

Comment on the financing obtained. Feel free to explain what the funding has funded:

AI initial API cost, salaries, legal and regulatory consulting.

FOUNDER 1

First name:	Adrian Peter
Last name:	Radomski
Address:	Cortex Park Vest 11, 2.4.
Zip code:	5230
City:	Odense M
E-mail address:	adra@mmmi.sdu.dk
Phone number:	22222337
Country:	Denmark

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Gender:	Male
Citizenship/Nationality:	Poland
Highest completed level of education:	Ph.d. og forskeruddannelser
From which educational institution did you graduate?:	University College Syddanmark - University College South Denmark
Name of education:	PhD Health Informatics and Technology
Graduation date or expected date:	15-01-2025
State the percent of ownership of the company for this founder:	45,0%

BUDGET

PROJECT

Amount applied for incl. overhead:	430.000 kr.
Monthly fee total:	330.000 kr.
Development costs total:	100.000 kr.

ATTACHMENTS

PERSONAL INFORMATION

PERSONAL INFORMATION

FOUNDERS

Number of founders: 1

FOUNDER 1

First name:	Adrian Peter
Last name:	Radomski
E-mail:	adra@mmmi.sdu.dk

STATEMENTS

APPLICATION FORMALITY

I have read and accepted the terms of use of the electronic application system (set out in the guidelines):



PERSONAL FORMALITIES

I am eligible to apply on behalf of the company:



I / we work full time on the business idea in the course:



I / we do not receive other public grants for what I apply for in Innofounder:



I / we are aware of the requirement for relevant work and residence permits to be able to work on the idea in Denmark in the event that I / we are foreign applicants from EU/EEA countries and countries outside the EU/EEA:



BUSINESS FORMALITIES

A) The applicant is not part of a company



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structure where an underlying company with a controlling influence is more than 5 years old.:

B) The applicant does not have more than 50 employees: ☒

C) The applicant does not have an annual turnover or balance sheet of more than EUR 10 million in the most recent audited financial statements: ☒

D) The applicant also fulfill A) and B) above, when the applicants companies, which seen in relation to the applicant should be regarded as associated or partner companies, as defined the the Comission's reccomendation concerning the definition of SMEs (2003/361/EF) are included: ☒

E) Public bodies do not control 25% or more of the Applicant's capital and / or voting rights: ☒

F) Applicant is not listed, ie. is not listed on a stock exchange's official list: ☒

G) The applicant was registered with the Danish Business Authority no more than 3 years ago at the time of the application: ☒

H) The applicant has not taken over the activity of another company, which at the time of the transfer either i) formed the basis for turnover with the transferor or ii) could realistically be expected to form the basis for turnover within a period of no more than 36 months: ☒

I) Applicant has not distributed profits: ☒

J) Applicant did not arise through a merger with a company that did not meet point F-I above: ☒

K) The applicant has not received an order for repayment of aid granted by the Danish state and has declared it illegal and incompatible with the Internal Market by the EU Commission, which the applicant has not yet complied with: ☒

L) The applicant is not an undertaking in difficulty regarding the definitions in GBER article 18, litra a-e: ☒

1. CV - Founder 1 CV

ADRIAN PETER RADOMSKI

PHD IN ENGINEERING SCIENCE & CTO/CO-FOUNDER, MARINA HEALTH APS

radomski.adr@gmail.com

+45 22 22 23 37

02/25/1995

Education

2025	Doctor of Philosophy (PhD) in Engineering Science, University of Southern Denmark, Odense, Denmark
2019	Master of Science in Electrical Engineering, Technische Universität München, Germany
2017	Bachelor of Science in Electrical Engineering, Technische Universität München, Germany

Academic Positions

2025 -	Postdoctoral Researcher, Health Informatics & Technology, Mærsk McKinney Møller Institute, University of Southern Denmark
2021 - 2025	PhD Student, University of Southern Denmark, Odense, Denmark

Industrial Experience

2019 - 2020	Engineer, Siemens, Princeton, NJ, United States
2018 - 2019	Student Employee, Siemens, Munich, Germany
2016 - 2017	Student Employee, Stemmer Imaging, Puchheim, Germany

Key Publications & Research Impact

2024 On-Road Evaluation of Unobtrusive In-Car Respiration Monitoring. A Radomski, D Teichmann. *Sensors* - Direct application to Marina's vital signs monitoring technology.

2024 Reflective PPG Sensor Measures Heart Rate Through Clothing During Driving. A Radomski, M Arai, K Oshima, N Oneda, A Ueno, D Teichmann. *IEEE Sensors Journal* - Breakthrough in non-invasive monitoring.

2023 Eliminating the Trade-off Between Resolution and Sampling Rate in Magnetic Induction Based Cardiorespiratory Sensors. A Radomski, D Teichmann. *IEEE EMBC 2023* - Core sensor technology expertise.

International Research Collaboration

2023	Research collaboration, Tokyo Denki University, Japan - Advanced sensor development
2018	Academic exchange, Nanyang Technological University, Singapore

August 28, 2025

2. Diplom - Founder 1

Det digitale ph.d.-bevis fra SDU

Ph.d.-beviser fra SDU udstedes pr. januar 2020 digitalt som PDF-dokument, hvor dekanens og ph.d.-skolens underskrift er indsat elektronisk.

Ved at klikke på verifikationslinket i sidehovedet, kan beviset tilgås i original på SDUs server.

Eksempler på gyldige ph.d.-beviser

Eksempler på hvordan digitale beviser fra SDU er udformet og valideres, kan ses på www.sdu.dk/tek/phddiploma

Hvad indeholder et digitalt ph.d.-bevis udstedt af Det Tekniske Fakultet?

- Forklæde med beskrivelse af det digitale bevis
- Bevisforside med gradserklæring
- Resultatoversigt
- Endelig bedømmelse (Separat dokument)

The digital PhD diploma at SDU

PhD diplomas at SDU are issued electronically as per January 2020. The diploma is issued as a PDF-document, with the signatures of the dean and the PhD school inserted electronically.

By following the verification link in the page header, the original diploma can be accessed on SDU's server.

Examples of valid PhD diplomas

Examples of diplomas and information on how to determine the validity of a diploma can be found at www.sdu.dk/tek/phddiploma

What constitutes the digital PhD diploma from the Faculty of Engineering

- A description of the digital diploma
- Degree certificate
- Transcript of study records
- Final evaluation (Separate document)

UNIVERSITY OF SOUTHERN DENMARK
HAS AWARDED

Adrian Peter Radomski

Civil Reg. No. 250295-3179

the PhD degree

in recognition of academic achievement in the area of

ENGINEERING SCIENCE

as demonstrated by a thesis entitled

Unobtrusive In-Car Vital Signs Monitoring

University of Southern Denmark, 5 February 2025


Henrik Bindslev, Dean

SYDDANSK UNIVERSITET
HAR TILDELT

Adrian Peter Radomski

CPR-nr. 250295-3179

Ph.d.-graden

I anerkendelse af videnskabelige kvalifikationer inden for

INGENIØRVIDENSKAB

godtgjort ved afhandlingen

Unobtrusive In-Car Vital Signs Monitoring

Syddansk Universitet, den 5. februar 2025


Henrik Bindslev, Dekan

Navn, Name

Adrian Peter Radomski

CPR-nr., Civil reg. No.

250295-3179**BILAG TIL PH.D.-BEVIS***Supplement to PhD certificate***I PH.D.-UDDANNELSEN INDGÅR FØLGENDE ELEMENTER:***The following elements are included in the PhD programme:*

KURSER: COURSES:	ECTS-points Credits
Getting Started on Your Teaching, SDU	2
English Grammar in Context, SDU	2
Collaborative Research and Participatory Methodologies in Educational Practices, SDU	0,5
Young Researcher Entrepreneurship Academy, DTU	2,5
Advances in Operations and Supply Chain Management, AAU	4
The Global PhD Platform Masterclass, SDU	5
Sustainable Development Goals and Global Production Systems: Exploring New Research Opportunities, SDU	5
Litteraturreport, SDU <i>Literature Report, SDU</i>	6
Konferencedeltagelse: IEEE EMBC 2023, International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), Sydney, Australien <i>Conference Participation: IEEE EMBC 2023, International Conference of the IEEE Engineering in Medicine & Biology (EMBC), Sydney, Australia</i>	3

Verification-link: <https://verify.sdu.dk/feafbc30-42c4-4e65-b7d5-881b0f6c53d8>

Navn, Name

Adrian Peter Radomski

CPR-nr., Civil reg. No.

250295-3179

BILAG TIL PH.D.-BEVIS

Supplement to PhD certificate

MILJØSKIFTEOPHOLD:

STUDIES AND RESEARCH IN ANOTHER RESEARCH ENVIRONMENT:

Japan, Tokyo, Tokyo Denki University, fra d. 15. februar 2023 til d. 5. maj 2023.

Japan, Tokyo, Tokyo Denki University, from 15 February 2023 to 5 May 2023.

VIDENFORMIDLING:

DISSEMINATION OF SCIENTIFIC KNOWLEDGE:

I løbet af ph.d.-uddannelsen er der gennemført vidensformidling svarende til minimum 300 timer.

During the PhD programme teaching activities or other forms of knowledge dissemination equivalent to 300 hours have been performed.

Syddansk Universitet, den 5. februar 2025

På ph.d.-skolens vegne
On behalf of the PhD School



Pia Ladefoged

Side 2

Dette er et digitalt eksamensbevis. Beviset kan kun valideres elektronisk. Se mere på www.sdu.dk/tek/phddiploma
This is a digital diploma. The diploma can only be verified digitally. See www.sdu.dk/tek/phddiploma for further information

3. Letters of Intent 1

Michael Stig

Head of Marine Standards, DFDS A/S

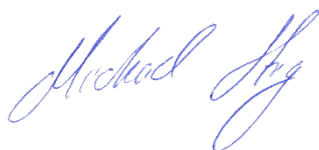
To whom it may concern,

DFDS A/S is pleased to express our support for the MARINA project – a digital maritime health record and decision support tool aimed at enhancing the quality and efficiency of medical care at sea. We see significant potential in this kind of solution to improve health-related decision-making onboard, reduce unnecessary medical evacuations, and contribute to a safer and more resilient maritime workforce.

In our view, MARINA could become a valuable standard across the industry by enabling structured, real-time medical assessments, supporting shipboard personnel, and improving communication with shore-based medical advisors. We believe this aligns well with the growing emphasis on digitalisation and crew welfare in modern shipping operations.

We look forward to following the development of MARINA and are interested in implementing MARINA when the product has been finalized.

Sincerely,
Michael Stig Head of Marine Standards

A handwritten signature in blue ink, reading "Michael Stig".

DFDS A/S
Marmorvej 18
2100 Copenhagen O
Denmark

M: +45 2944 3220
www.dfds.com

4. Letters of Intent 2

Bugge Marx Hansen
Kvaglundvej 21
6705 Esbjerg Ø, Denmark
Phone
buggemarx@gmail.com

To Peter Biesenbach

I am writing to give my opinion of the current version of the “Maritime Medical Record Assistant”. These considerations are based on my experience as a medical officer or Master on various vessels in the last 25 years. After a demonstration and review of the system’s capabilities, my conclusion is the following:

The “Maritime Medical Record Assistant” developed by your team is an innovative solution which addresses several important needs within the maritime industry and potentially other remote operations. It could significantly enhance the medical care available on vessels, and improve the safety of acutely sick seafarer.

Areas of Interest:

1. Multilingual Speech Recognition: The ability to transcribe patient assessments in multiple languages is invaluable for our diverse crew members, preventing the common errors rooted in communication and translation between different nationalities.
2. AI-Powered Medical Record Generation: The automatic extraction of medical information from verbal assessments into structured records significantly improves documentation workflows. For seafarers with dyslexia or partly dyslexia, I can only see benefits within this program.
3. Digital Form Management: The intuitive interface for reviewing and completing medical records ensures no critical information is missed.
4. Emergency Assessment Capabilities: The urgency assessment and “red flag” notification features align with safety protocols, would enhance our risk management, ensuring that a physician is contacted right away in critical situations which otherwise are missed.
5. Offline Accessibility: The capability to generate reports without constant internet connectivity is essential for our vessels operating with limited internet access.

Potential Implementation:

As an experienced medical officer and potential end-user of the “Maritime Medical Record Assistant” in the case of a medical emergency on board, I anticipate that implementation of this system would result in:

- Improved quality and consistency of medical reports
- More efficient generation of medical reports
- Better communication between shipboard personnel and onshore medical support

- Quicker diagnosis and treatment
- Better medical outcomes through standardized assessment protocols
- Reduced administrative burden on our medical officers

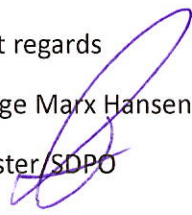
Closure comments:

I look forward to see the final version of the “Maritime Medical Record Assistant” and in my opinion, I can only see pros within the system’s capabilities.

Best regards

Bugge Marx Hansen

Master/SDPO



Line Emilie Lilholm Laugesen
Finsensgade 35
6700 Esbjerg, Denmark
Line.Emilie.Lilholm.Laugesen@rsyd.dk

23th of April 2025
RE: Letter of Interest for MARINA

To whomever it concerns

I hereby summarize my consideration of the MARINA - the Maritime Medical Record Assistant as presented this week. My words are based on my experience as an emergency nurse, researcher and general manager at Radio Medical Denmark, the 2nd largest provider of tele-medical assistance service to ships in Europe. After independently assessing the system, I conclude as follows:

MARINA is an innovative solution with the potential to improve health care within the maritime industry as well as platforms and similar conditions.

Considerations

The following features could be particularly helpful:

Given the challenges medical officers on vessels face when communicating with a diverse crew, a multilingual assistant trained specifically for maritime health issues could be helpful to gather information from patient. There is always a risk for loss of information due to language barriers and cultural differences. The option for speech recognition allows the patient to report the symptoms in the language they are most comfortable with.

The standardized process of collecting information based on the chief complaint and the suggested examinations could help the medical officer in collecting all relevant information. This could particularly help officers who do not have a lot of experience in the examination of acutely sick patients.

The automated process of record generation could be useful for work flow on the ship, because it appears to be quick and efficient. I particularly like the inclusion of a Red Flag system, which informs the medical officer if immediate contact to a maritime physician is required.

It is my understanding that the underlying idea of MARINA is not to replace or delay the human interaction between the patient, the medical officer and the maritime physician, but to guide medical officers in making a comprehensive medical report.

Suggestions

To investigate the potential of MARINA - Maritime Medical Record Assistant, a formal pilot trial evaluating the tool has to be conducted. This could be done on shore with real sailors, in collaboration with the training center on Fanø.

Please consider this letter as formal confirmation of my serious interest in participating in the research projects related to MARINA.

Sincerely,

A handwritten signature in black ink that reads "Line Emilie Lilholm Laugesen". The script is cursive and fluid, with the first name "Line" and last name "Laugesen" being more prominent.

Line Emilie Lilholm Laugesen
RN, MScPH

5. Letters of Intent 3

Karsten Kristensen
3F Transport
Email: karsten.kristensen@3f.dk

To whom it may concern,

3F's Transport, being a long-standing advocate for the health and safety of maritime workers, would like to express our strong endorsement of the MARINA Health AI initiative.

Seafarers face unique healthcare challenges, including isolation from land-based medical services, language barriers, and delays in receiving appropriate treatment. MARINA offers a promising and innovative digital solution to address these issues—leveraging artificial intelligence and speech recognition to support remote medical decision-making onboard vessels.

From the perspective of worker protection and occupational health, a system like MARINA has the potential to:

- Improve early identification of serious illness,
- Support onboard crew and captains in emergency situations,
- Reduce miscommunication in medical triage and teleconsultation,
- And ultimately enhance outcomes and safety for all seafarers.

We view MARINA as aligned with international efforts to modernize maritime safety standards, and we are pleased that the project is rooted in a Danish research and healthcare context, with responsible public-sector backing.

Best regards
Karsten Kristensen
3F Transportgruppen

6. Slidedeck

maritime healthcare crisis



Language Barriers

Multilingual crews create dangerous miscommunication during emergencies



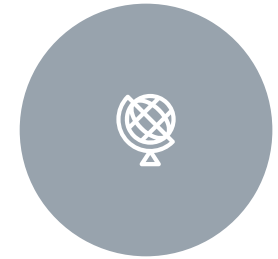
Poor Record Keeping

Inconsistent documentation causes treatment delays



Limited Medical Training

Officers lack skills for complex remote cases



Cultural Differences

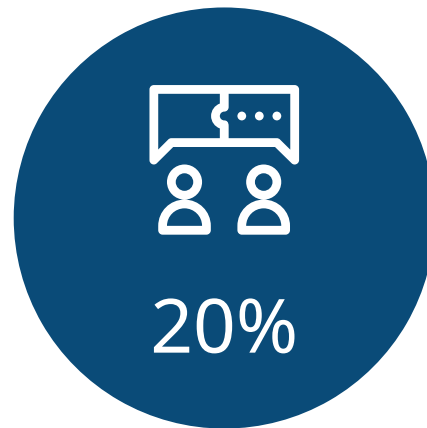
Varied health attitudes hinder accurate diagnosis

market opportunity



ships

Divert annually



Of them

Are preventable



cost

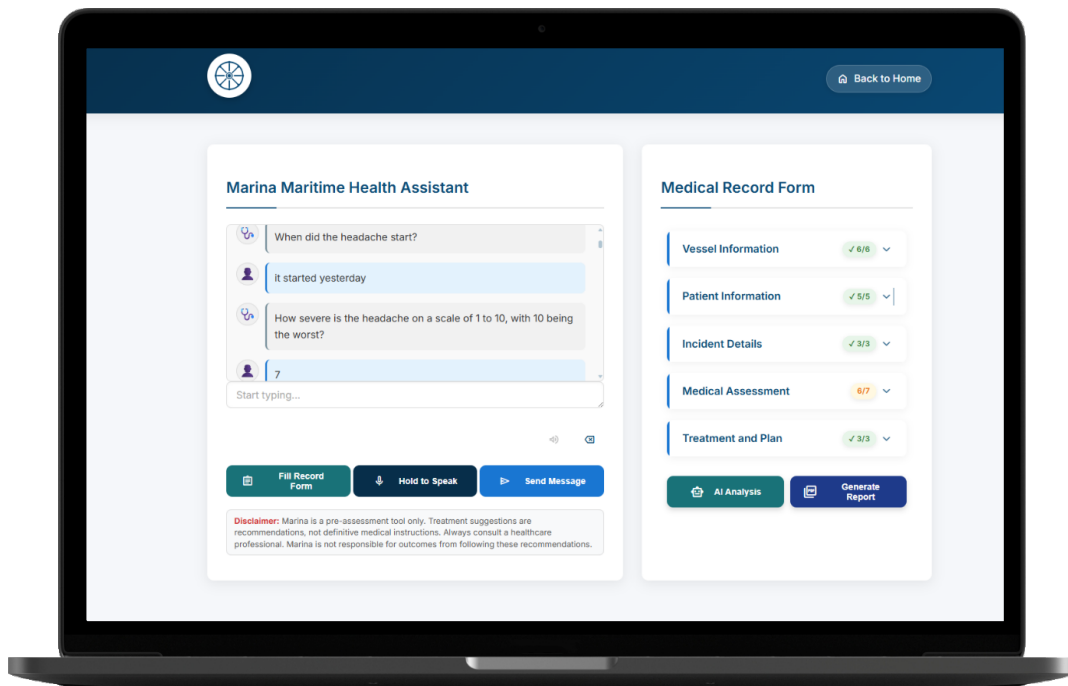
Of diversion



overall

Industry costs

AI Medical Assistant for MARITIME ENVIRONMENTS



Marina provides structured medical assessment, multilingual support, and AI-assisted documentation when professional care is miles away.

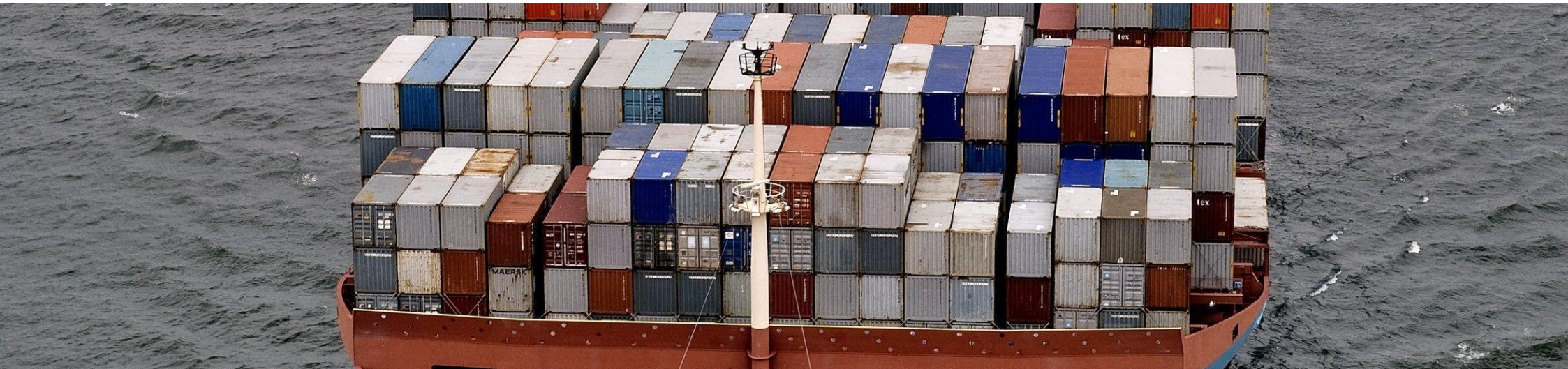
+100
languages

English, Filipino,
Russian & more

offline
operation

Everywhere in the
world

About Our Solutions



Voice-first interviews

Natural AI conversations for in-depth assessment

MULTILINGUAL SUPPORT

Break barriers with 30+ languages

URGENCY ASSESSMENT

Red flag warnings for critical conditions

STANDARDIZED DOCUMENTATION

Auto-generate compliant medical records

Outcomes for shipping companies



COST REDUCTION

Minimize expensive diversions and evacuations with AI-guided medical assessments



STANDARDIZED RESPONSE

Ensure consistent medical protocols across your entire fleet regardless of location



LEGAL PROTECTION

Demonstrate duty of care with compliant documentation and reduce liability exposure



SMARTER ESCALATION

Help officers confidently determine when to call for help and when to monitor